Alexander Hölzemann

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Github | LinkedIn | Google Scholar (h-index: 10)

SKILLS

ML & Modeling: Keras/TensorFlow, PyTorch (basic), scikit-learn, self-supervised, supervised and unsupervised learning
Data & Scientific Computing: NumPy, Pandas, SciPy, xarray, Zarr, Dask
Visualization: Matplotlib, Plotly
Compute & Cloud: HTCondor, HTC/HPC environments, Docker (basic), AWS (S3, boto3)
Experimentation: Weights & Biases, Optuna
Engineering & Collaboration: Git, Jupyter, Slack/Mattermost, LaTeX

Additional / Prior: SQL, Unix/Bash, Java, JavaScript/TypeScript, C/C++, R, Dart, Flutter, Angular,
Languages: German (native), English (C2), Spanish (C1)

PROFESSIONAL EXPERIENCE

2026 **Postdoctoral Researcher | Department of Computer Science | University of Colorado Boulder, Boulder, USA |** *September 2024 - Present*

- Engineered a self-supervised learning pipeline to separate biological acoustic signatures from background signals and support fish species and schooling-pattern classification.
- Developed Autoencoder/Decoder and BYOL models and implemented multilevel BGMM clustering with sub-clustering to isolate biological signatures from background.
- Architected cloud-native deep learning pipelines to stream NOAA petabyte-scale water-column sonar data from AWS S3 using Zarr and Xarray, eliminating local storage limits.
- Engineered a scalable machine learning pipeline optimized for distributed execution and high-throughput processing on a parallelized HTC cluster.
- Contributed to the Project Pythia OSDF-Cookbook, developing reproducible Jupyter Notebook workflows for streaming NOAA Water-Column Sonar Data via the Open Science Data Federation (OSDF) to enable cloud-based, scalable scientific analysis.

2024 **Scientific Researcher and Ph.D. Candidate | Department of Ubiquitous Computing | University of Siegen, Germany |** *May 2018 – August 2024*

- Developed machine and deep learning pipelines and models for Human Activity Recognition (HAR)
- Investigated transfer learning behavior for wearable-sensor HAR (Project: Digging Deeper) and implemented end-to-end evaluation pipelines.
- Developed a cross-correlation algorithm to synchronize data streams from multiple wearable IMUs, aligning timestamps with millisecond precision.
- Architected full-stack data collection tooling (Project: ActiVATE), building a cross-platform mobile application (Flutter) alongside a backend and database to store participant data.
- Led study design and data collection for basketball activity recognition (Project: Hang-Time HAR), annotating over 50 hours of data and releasing a benchmark dataset via Zenodo.

PROFESSIONAL INTERNSHIPS & EARLY EXPERIENCE

2018 **Research Assistant** | University of Siegen, Germany | 2014 – 2018
Developed GIS-based AngularJS web applications for disaster relief (Project Kokos) and JavaEE-based tools for competency assessments (Project ComProFITS).

2016 **Working Student (Mobile & Web Dev)** | Krombacher Brewery, Germany
Developed web and mobile applications for inventory management and workplace optimization using barcode scanning.

2015 **IT Internship** | Oiltanking Ebytem S.A., Argentina
Integrated and customized the GLPI platform for IT asset management and monitoring.

2015 **Working Student (IT Support)** | SMS Siemag AG, Germany | 2010 – 2015
Provided technical support for hardware, mobile devices, and industrial IT infrastructure.

EDUCATION

2024 **Ph.D. in Computer Science**, Department of Ubiquitous Computing, University of Siegen, 57076 Siegen, Germany, Final Grade: Magna Cum Laude

2018 **Master of Science in Computer Science (Major in Software Architecture)**, University of Siegen, 57076 Siegen, Germany - Final Grade: 2.0 (German), 3.3 (US-GPA)

2013 **Bachelor of Science in Computer Science (Major in Media Studies)**, University of Siegen, 57076 Siegen, Germany - Final Grade: 2.7 (German), 2.7 (US-GPA)

LEADERSHIP & MENTORING

Project & Event Leadership: Co-General Chair for international workshops (WellComp 2023/2024), overseeing committee appointments, budget, and on-site operations.
Team Coordination: Recruited and managed international teams of student volunteers for large-scale conferences (UbiComp/ISWC).
Technical Mentoring: Designed and delivered a tutorial on Deep Learning for HAR, covering end-to-end sensor data preprocessing, noise filtering, and class balancing for production-ready model training.
Education: Guided students in applying deep learning methods to practical projects and supervised scientific writing and research synthesis.

PUBLISHED MANUSCRIPTS

Peer-reviewed journals, conferences or workshops.
Google Scholar h-Index 10, 402 Citations (as of May 2026)

2024 Wu, S., **Hoelzemann A.**, Bock M., Van Laerhoven K., Plötz T., Adams A.. Basketball shooting performance analysis using multi-modal wearable and mobile sensing in semi-naturalistic settings. 2024 IEEE 20th International Conference on Body Sensor Networks (BSN). IEEE DOI

2024 Dang T., Gashi S., Spathis D., **Hoelzemann A.**, WellComp 2024: Seventh International Workshop on Computing for Well-Being. Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing. DOI

2024 **Hoelzemann, A.**, and Van Laerhoven K.. "A matter of annotation: an empirical study on in situ and self-recall activity annotations from wearable sensors. Frontiers in Computer Science 6, DOI.

2024 **Hoelzemann, A.**, Bock M., & Van Laerhoven, K. Evaluation of Video-Assisted Annotation of Human IMU Data Across Expertise, Datasets, and Tools. In *2024 IEEE International Conference on Pervasive Computing and Communications Workshops and Other Affiliated Events (PerCom Workshops),* DOI.

2023 **Hölzemann, A**. Improved training approaches for embedded learning with heterogeneous sensor data **(Dissertation**) DOI.

2023 **Hoelzemann, A.**, Bock, M., Valladares Bastías, E. A., El Ouazzani Touhami, S., Nassiri, K., & Van Laerhoven, K. A Data-Driven Study on the Hawthorne Effect in Sensor-Based Human Activity Recognition. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing* (pp. 486-491). DOI

2023 Gashi, S., Spathis, D., Dang, T., & **Hoelzemann, A.** WellComp 2023: Sixth International Workshop on Computing for Well-Being. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing* (pp. 791-794). DOI

2023 **Hoelzemann, A.**, Romero, J. L., Bock, M., Laerhoven, K. V., & Lv, Q. Hang-Time HAR: A Benchmark Dataset for Basketball Activity Recognition Using Wrist-Worn Inertial Sensors. *Sensors*, *23*(13), 5879. DOI

2023 **Hoelzemann, A.**, & Van Laerhoven, K. A Matter of Annotation: An Empirical Study on In Situ and Self-Recall Activity Annotations from Wearable Sensors. *arXiv preprint and currently under review for Springer Personal and Ubiquitous Computing*. DOI

2022 Bock, M., **Hoelzemann, A.**, Moeller, M., & Van Laerhoven, K. Investigating (re) current state-of-the-art in Human Activity Recognition Datasets. *Frontiers in Computer Science*, *4*, 119. DOI

2022 Mähs, M., Pithan, J. S., Bergmann, I., Gabrys, L., Graf, J., **Hoelzemann, A.**, ... & Teti, A. Activity Tracker-Based Intervention to Increase Physical Activity in Patients with Type 2 Diabetes and Healthy Individuals: Study Protocol for a Randomized Controlled Trial. *Trials*, *23*(1), 617. DOI

2022 Van Laerhoven, K., **Hoelzemann, A.**, Pahmeier, I., Teti, A., & Gabrys, L. Validation of an Open-Source Ambulatory Assessment System in Support of Replicable Activity Studies. *German Journal of Exercise and Sport Research*, *52*(2), 262-272. DOI

2022 **Hoelzemann, A.**, Pithan, J. S., & Van Laerhoven, K. Open-Source Data Collection for Activity Studies at Scale. In *Sensor-and Video-Based Activity and Behavior Computing: Proceedings of 3rd International Conference on Activity and Behavior Computing (ABC 2021)* (pp. 27-38). Singapore: Springer Nature Singapore. DOI

2021 Bock, M., **Hoelzemann, A.**, Moeller, M., & Van Laerhoven, K. Tutorial on Deep Learning for Human Activity Recognition. *arXiv preprint*. DOI

2021 Bock, M., **Hoelzemann, A.**, Moeller, M., & Van Laerhoven, K. Improving Deep Learning for HAR with Shallow LSTMs. In *Proceedings of the 2021 ACM International Symposium on Wearable Computers* (pp. 7-12). DOI ☆ **BEST PAPER AWARD**

2021 Wegmeth, L., **Hoelzemann, A.**, & Van Laerhoven, K. Detecting Handwritten Mathematical Terms with Sensor Based Data. *arXiv preprint.* DOI

2021 Ramachandra, S., **Hoelzemann, A.**, & Van Laerhoven, K. Transformer Networks for Data Augmentation of Human Physical Activity Recognition. *arXiv preprint.* DOI

2021 **Hoelzemann, A.**, Sorathiya, N., & Van Laerhoven, K. Data Augmentation Strategies for Human Activity Data Using Generative Adversarial Neural Networks. In *2021 IEEE International Conference on Pervasive Computing and Communications Workshops and Other Affiliated Events (PerCom Workshops)* (pp. 8-13). IEEE. DOI

2020 **Hoelzemann, A.**, & Van Laerhoven, K. Digging Deeper: Towards a Better Understanding of Transfer Learning for Human Activity Recognition. In *Proceedings of the 2020 ACM International Symposium on Wearable Computers* (pp. 50-54). DOI

2019 **Hoelzemann, A.**, Odoemelem, H., & Van Laerhoven, K. Using an In-Ear Wearable to Annotate Activity Data across Multiple Inertial Sensors. In *Proceedings of the 1st International Workshop on Earable Computing* (pp. 14-19). DOI

2019 Odoemelem, H., **Hoelzemann, A.**, & Van Laerhoven, K. Using the eSense Wearable Earbud as a Light-Weight Robot Arm Controller. In *Proceedings of the 1st International Workshop on Earable Computing* (pp. 26-29). DOI

2018 **Hoelzemann, A.**, & Van Laerhoven, K. Using Wrist-Worn Activity Recognition for Basketball Game Analysis. In *Proceedings of the 5th International Workshop on Sensor-based Activity Recognition and Interaction* (pp. 1-6). DOI

PUBLISHED DATASETS

2023 **Hoelzemann A.**, & Van Laerhoven K. Self-Annotated Wearable Activity Data (v. 0.1). Zenodo. DOI

2023 **Hoelzemann, A.**, Romero, J. L., Bock, M., Laerhoven, K. V., & Lv, Q. Hang-Time HAR: A Benchmark Dataset for Basketball Activity Recognition using Wrist-worn Inertial Sensors. In MDPI Sensors: Bde. Sensors 2023, 23(13), 5879 (Inertial Measurement Units in Sport). Zenodo. DOI

ACADEMIC VISITS

2023 **Visiting Scholar, Department of Computer Science, University of Colorado Boulder, Boulder, CO, USA.**
March & Juli

- Collaborated with Prof. Qin Lv's research team at the University of Colorado Boulder on the Hang-Time HAR dataset, leading study design and data collection sessions. Dataset featured in published MDPI Sensors journal article.

2019 **Visiting Scholar, Center for Ubiquitous Computing, University of Oulu, Finland.**
December

- Established an Erasmus+ academic cooperation for student and faculty exchanges between the Center for Ubiquitous Computing at the University of Oulu and the Department of Ubiquitous Computing at the University of Siegen.

TEACHING

2024 **Lecturer,** Seminar Data Science**, University of Siegen, Germany**
Summer Term 2020 - present

- Teaching independent writing of a scientific paper, including the ability to apply algorithms and concepts of machine learning to practical programming projects, access material on a given topic from literature databases and other sources, read and understand and prepare original English literature.

2021 **Lecturer,** Recent Advanced in Machine Learning**, University of Siegen, Germany**
Summer Term 2019 - Winter Term 2020/2021

- This module presents recent advances in machine learning in different fields of data sciences including imaging, vision, graphics, mechatronics, and sensorics. It addresses advanced techniques in the fields of machine learning, deep learning and artificial intelligence, with a particular focus on recent research papers, novel application areas and open questions in the aforementioned fields.

2020 **Teaching Assistant,** Ubiquitous Computing, **University of Siegen, Germany**

Summer Term 2018 - Winter Term 2019/2020

- This lecture gives an overview on relevant concepts and technologies, such as the history of ubiquitous computing and underlying visions, embedded systems and cyber-physical systems, mobile computing, wearable computing, and wireless sensor networks.

ACADEMIC SERVICES

2024 **Scientific Reviewer**

since 2018 - present

- Reviewed manuscript submissions as an expert referee for major scientific publishers including ACM, IEEE, Springer, and Frontiers. Provided rigorous technical evaluation across conferences, journals and workshops on activity recognition topics, such as UbiComp/IMWUT and ISWC, IEEE Percom, Activity and Behavior Computing (ABC), EarComp, WellComp, HASCA, IWOAR.

2023 **Co-General Chair** at WellComp 2023 (Cancún, Mexico)

- General Chairs are the main organizers of events such as conferences, workshops, and symposia, They appoint the chairs of other committees that make up the conference committee and chair the in-person event.

2023 **Co-Web Chair** at UbiComp/ISWC 2023 (Cancún, Mexico)

- Overseeing end-to-end design and content upkeep of the official conference website using WordPress. Managed site sections including calls for papers, workshop/tutorial proposals, speaker details, conference schedules/logistics, local attractions, and travel information.

2021 **Co-Chaired** a Tutorial on Deep Learning for Human Activity Recognition, UbiComp/ISWC 2021 (Virtual)

- Designed and conducted a session on data preprocessing for Human Activity Recognition. Covered techniques including noise filtering, up/down sampling, missing data imputation, and class balancing to ready sensor streams for deep learning.

2020 **Co-Student Volunteer Chair** at UbiComp/ISWC 2020 (Virtual)

- Managed recruitment, onboarding, and coordination of student volunteers supporting critical event operations. Assigned volunteers to registration, session coordination, technical support, and speaker roles based on multifaceted conference needs across 5 days. Ensured positive attendee experiences through prompt issue resolution.

2017 **Founding member** of the Exchange Students Siegen e.V, Siegen – Germany

- The Exchange Students Siegen e.V. is part of the Erasmus Student Network (ESN). It is a non-profit international student organization. As a local section of ESN Germany and ESN International, they care for all incoming international students in Siegen.